AIRGUARD INDIA 2.0

AirGuard India — final execution report, 20 October 2026

DATE	20 October 2026
SOURCE	reports/_final_execution_report.md
REPOSITORY	https://github.com/nishaids/airguard-india

AirGuard India — completion report, 20 October 2026

Supersedes the 13 October checkpoint (preserved in git history). Every number below comes from a committed artefact named beside it. A working deployment does not show forecast accuracy, and the two are reported separately.

1. Executive summary. Production is updated and verified; the research is **not** complete. The primary CAMS experiment cannot run until the archive holds enough months: 10 of 36 files are verified and the first protocol-eligible fold needs 10 more. Acquisition was re-authorised by the author today and is running. No model was selected or locked, the sealed final test was not touched, and the paper is unchanged.

2. Repository checkpoint. Branch `research/station-calibration`, pushed to `github.com/nishaids/airguard-india`. Working tree clean at each commit; no force-push, no history rewrite.

3. Earlier Claude Code and Codex work recovered. All preserved: measured dataset (v3.1), locked protocol, six-page paper snapshot, exploratory Severe, uncertainty, warning and survivorship extensions, guarded primary runner, receipt provenance, prospective ledger, verified 13 October deployment. Nothing was redone.

4. Data sources. `DATA_SOURCE_REGISTER.md` (generated): ten sources classified OBSERVED / ARCHIVED_OPERATIONAL_FORECAST / LIVE_FORECAST / MODELLED_OUTPUT; no reanalysis used. Station targets are CPCB-network monitors through OpenAQ, not official CPCB data. The official data.gov.in feed has 0 successful and 187 failed recorded ingestions; today its host answered but returned HTTP 502 after about 60 s.

5. CAMS acquisition. `CAMS_ARCHIVE_STATUS.md`: 10/36 verified, 2 running, 24 to submit, 0 duplicates, 0 unresolved failures. The June request pending since 13 October had expired on the server (HTTP 404) and was retired with its ID kept; the worker now downloads each result at once.

6. Station matching. `CAMS_STATION_MATCHING_REPORT.md`: 18,540 station-days, 180 stations, 100 cities, 103 dates, 11,634 exact and 7 censored Severe targets, **0 availability violations**. Rebuilt today with identical hashes. Nearest 0.4-degree cell; coastal cells not yet audited.

7. Development model comparison. Not run: preflight `BLOCKED_NO_ELIGIBLE_FOLDS`. An exploratory baseline diagnostic was run instead, fitting and selecting nothing (`RAW_CAMS_DEVELOPMENT_DIAGNOSTIC.md`): on 9,510 development station-days raw CAMS misses measured AQI by 93.4 and runs 82.6 high, against 25.7 for persistence (paired difference +67.7, 95% block interval 60.8 to 77.5), because CAMS ozone averages 2.6 times the measured value. Folds 1-4 can never meet the locked 120 valid training days per station; fold_05 needs CAMS through 2025-11-16.

8. Model selection. NOT READY. No `MODEL_SELECTION_LOCK` (correctly). No rule was relaxed.

9. Severe events. Exploratory, non-CAMS only (`DEVELOPMENT_EXTENSION_AUDIT.md`): no tail learner dominates; Severe kept as lower-bound censored. Primary evaluation pending.

10. Uncertainty. Exploratory only; marginal coverage hides poor coverage in polluted categories (pilot: 57-75% of Severe days inside nominal 90% intervals). Production shows no interval ("not yet available").

11. Warning policy. Exploratory operating points only; no policy called optimal; none deployed.

12. Geographic robustness. Non-CAMS temporal and geographic holdouts in the paper; CAMS holdouts pending. No nationwide generalisation claim.

13. Survivorship. Frozen-cohort sensitivity preserves the main qualitative findings; retired monitors were never acquired (`STATION_SURVIVORSHIP_SENSITIVITY.md`).

14. Literature and novelty. `FINAL_NOVELTY_AND_PRIOR_ART_AUDIT.md`: 48 of 56 references verified by Crossref, arXiv or the official page; a logged Crossref search (10 queries, 190 records, 13 added); 28-study matrix; gap and claim matrices. Verdict: every method has prior art; no first-ever or state-of-the-art claim is supported; the combination is a defensible design that becomes a contribution only after the primary experiment.

15. Public-impact functionality. Measured/modelled labels, freshness labels, tomorrow forecast with issue time, pollutant breakdown, general exposure guidance with disclaimer, nearest-station context, forecast ledger. Today: the live track record now publishes error figures only with ≥30 checked forecasts over ≥7 days (it had shown one-day anecdotes such as a +249.5 bias), and stale data is no longer cached as current. A second release states the raw model's measured accuracy on the page and warns when ozone leads the estimate.

16. Public-impact evaluation. `PUBLIC_IMPACT_EVALUATION_PLAN.md`: task-based protocol with pre-registered thresholds. Not run: needs ethics approval. No data collected.

17. Prospective verification. 108 fixed-time forecasts over 9 target days, 2 verified days, 0 availability violations; live product: 48 forecasts checked on 3 days. No skill claim.

18. Security. Secret audit 0 hits across 2,327 git objects, the working tree and 50 browser assets; CSP, HSTS, nosniff, frame-deny and permissions headers live; all database tables under RLS. Credentials once pasted in chat should be rotated by the author.

19. Backend tests. 681 passed (full suite); source-bound offline validation 677 collected, 672 passed, 5 skipped, 0 failed.

20. Frontend tests. 44 unit tests pass; lint, type check and production build pass.

21. Browser tests. Live home page checked at desktop and 375 px; track-record section verified after release. A first visit to an uncached place takes about 20 s with an honest loading state.

22. Performance. Cached summary responses 0.2 s; uncached about 3 s per request with a queued 20 s first build. No new infrastructure added.

23. GitHub. Pushed; remote head matches local at push time.

24. Deployment. `docs/DEPLOYMENT_COMPLETION.md`: two verified releases today, latest `dpl_5r2cg4ZKdAtgRMHo1TpoXh2653L9`; each passed 31 staged and 37 public checks. The first candidate was not promoted because staging exposed a cached-format hazard; fixed first.

25. Paper. PRESERVED. `paper/` unchanged this session.

26. Final test. SEALED. Code-level guards and tests refuse sealed rows. Approval not requested: no lock, no READY audit.

27. Remaining blockers. CAMS archive (ADS queue time; needs the laptop awake); ethics approval for the study; author acceptance of the literature audit; time for the prospective record; official CPCB feed (server-side 502).

28. Next human action. Keep the CAMS worker running until at least the November 2025 files verify (command in `CAMS_ARCHIVE_STATUS.md`); rotate the API keys and database password that were pasted into chat; decide on ethics submission for the comprehension study.

Gates (`results/research/research_readiness.json`): research 5/17, `COLLECTING_NOT_RESEARCH_READY`. Production gates for this release: all met.

Final execution audit — 20 October 2026

Branch `research/station-calibration`, pushed to GitHub. Production deployment `dpl_4YKByhjbQLA8Z7Tsi9U7PXSP44Bd`. Final test (10 July to 7 October 2026) sealed. A passing test suite or a working deployment is not evidence of forecast skill.

Recovered state. On arrival the tree was clean at `2c96805` (13–14 October work by earlier Claude Code and Codex sessions, all preserved): CAMS archive 8 of 36 files with new ADS submissions paused by the author, a 103-date development join, model selection NOT READY, a verified production deployment, the frozen six-page paper. The one pending ADS request (June 2025) had expired on the server (HTTP 404) before download.

COMPONENT	STATUS	EVIDENCE	REMAINING WORK	DEPENDS ON	PRIORITY	DONE WHEN
Repository and history	COMPLETE	Clean tree, logical commits, pushed `6f4da73` → later	—	—	P4	Pushed
Paper snapshot	COMPLETE	`paper/` unchanged this session (`git diff` empty)	Author submission decisions	Author	—	Unchanged
Data source register	COMPLETE	`DATA_SOURCE_REGISTER.md`, `data_source_manifest.json` (generated)	Regenerate after archive updates	—	P0	Generated from evidence
CAMS archive	PARTIAL	10 of 36 verified; author re-authorised 2026-10-20; worker running	26 files	ADS queue, laptop awake	P1	36 of 36 verified
CAMS-station matching	PARTIAL	`CAMS_STATION_MATCHING_REPORT.md`; join rebuilt with identical hashes; 0 availability violations	Grow with the archive; coastal audit	Archive	P1	Join covers the plan
Primary model comparison	BLOCKED	Preflight `BLOCKED_NO_ELIGIBLE_FOLDS`	Run declared comparison	fold_05 (CAMS through 2025-11-16)	P1	All declared experiments on identical rows
Model selection and lock	BLOCKED	No lock (correctly)	Select on development folds, freeze	Primary comparison	P1	Committed `MODEL_SELECTION_LOCK.md`
Severe-event research	PARTIAL	Exploratory non-CAMS extension	Primary severity evaluation	Winter folds	P1	Tail metrics on CAMS folds
Uncertainty	PARTIAL	Exploratory; no interval deployed	Primary Mondrian/CQR	Primary comparison	P2	Coverage by category with sizes
Warning policy	PARTIAL	Exploratory operating points	Cost function, primary evaluation	Primary comparison	P2	Conservative and balanced modes evaluated
Geographic robustness, survivorship	PARTIAL	Non-CAMS holdouts; frozen-cohort sensitivity	CAMS holdouts	Archive	P3	Leave-city/cluster-out on CAMS
Literature and novelty	PARTIAL	48 of 56 references verified; logged search; 28-study matrix; gap and claim matrices	Full texts of title-only studies; human acceptance	Author	P3	Accepted audit
Public-impact features	COMPLETE for this release	Honest track record, freshness fix, measured/modelled labels, guidance, ledger	Comprehension evidence	Ethics approval	P2	—
Public-impact evaluation	BLOCKED	`PUBLIC_IMPACT_EVALUATION_PLAN.md`	Run study	Ethics approval	P3	`results/user_study.json` from the protocol

COMPONENT	STATUS	EVIDENCE	REMAINING WORK	DEPENDS ON	PRIORITY	DONE WHEN
Prospective verification	PARTIAL	108 fixed-time forecasts, 2 verified days, 0 violations; live track record gated	Time	Elapsed days, station data	P3	≥30 verified days
Monitoring	PARTIAL	Status page, provider events, issuance heartbeat, freshness labels	Model-performance and coverage drift once a model is deployed	A deployed model	P4	Drift alerts with baselines
Backend tests	COMPLETE	681 passed (full); 672 passed, 5 skipped (source-bound offline)	—	—	P4	Green
Frontend tests	COMPLETE	44 unit tests; lint, types, build pass	—	—	P4	Green
Security	COMPLETE (configured-secret scope)	Secret audit 0/0/0 over 2,327 git objects, worktree, 50 browser assets; security headers live	Rotate credentials once pasted in chat	Author	P4	Rotated
Deployment	COMPLETE	`docs/DEPLOYMENT_COMPLETION.md` : 31 staged + 37 public checks	—	—	P4	Verified live
Final test	SEALED	Guards and tests; no lock	Author approval after READY	Lock	—	Scored once, after approval

Changes made in this session

1. Resumed CAMS acquisition under the author's approval: reviewed handling of the expired June result, authorised concurrency of two, prompt download. Tests added.
2. Live track record: error figures published only with ≥30 checked forecasts over ≥7 days; contract and tests enforce it; cached legacy cells handled.
3. Freshness: summaries built while a source refreshes are cached ≤30 s.
4. Data source register, CAMS-station matching report, literature verification and search, novelty and prior-art audit, public-impact plan, journal roadmap, limitations update.
5. Release staged, one candidate rejected on evidence, fixed, promoted and verified.

Not done, and why

- Primary CAMS experiment, model selection, lock, final-test request: the archive lacks the months the locked protocol needs. Nothing was relaxed to force a result.
- User study: needs ethics approval.
- Premium visual redesign: deliberately not prioritised over correctness while research gates are open.

Raw CAMS against station measurements — exploratory development diagnostic

Status: EXPLORATORY, NOT THE PRIMARY EXPERIMENT. No model was fitted, tuned or selected. Two baselines the locked protocol already defines were scored on the same station-days of the guarded development join, which ends before the sealed final test. Script: `scripts/explore_raw_cams_development.py` ; numbers: `results/research/extensions/raw_cams_development_diagnostic.json` .

Result

On **9,510 station-days** (21 February to 1 June 2025, exact targets only):

BASELINE	MAE (AQI)	BIAS	CATEGORY AGREEMENT	POOR-OR-WORSE DAYS MISSED
Raw archived CAMS, matched pollutants (B2)	93.4	**+82.6**	32%	55% of 572
Raw CAMS, all six pollutants (as the live site shows)	93.4	+82.6	32%	55%
Persistence, measured 24-hour index at the cutoff (B1)	25.7	+1.0	69%	51%

Raw CAMS minus persistence, paired MAE difference: **+67.7 AQI** (95% seven-day block-bootstrap interval 60.8 to 77.5, 91 dates).

Why

Ozone. CAMS ozone averaged 140 µg/m³ against 54 measured (2.6 times), so ozone set the CAMS index on 81% of station-days, against 15% in the measurements. PM2.5 was close on average (46 against 43); PM10 was higher (149 against 103). The bias was worst in the North (MAE 122, bias +113) and grew from February (MAE 62) to May (107). Matched and six-pollutant indices were identical on every station-day here, because NO_2, SO_2 and CO never

set the index.

The raw model is too high on ordinary days yet still misses over half of the Poor-or-worse days: it is mostly wrong about *which* days are bad, not uniformly high.

What follows, and what does not

- The live product shows a raw CAMS-based AQI. Since 20 October 2026 the page states these figures and warns when ozone leads the estimate. Nothing is corrected: a correction would be an unvalidated model.
- This is the strongest motivation yet for station calibration, and a reason to expect it to help. It is **not** evidence that calibration works: that needs the primary experiment on locked folds.
- One season only (no winter, few Poor-or-worse days). Persistence assumes a healthy feed with 3-hour latency.

CAMS operational archive status

Checkpoint: 20 October 2026 (IST). Supersedes the 13 October checkpoint, which is preserved in git history. Final test sealed.

Current manifest: 10 of 36 planned files verified, 2 running, 24 not yet submitted. Frozen plan SHA-256 `952325f75dfc11e773dda90cbfe4ec4e883cda045a30d03c17578980ababeb6a` (unchanged): 507 00 UTC operational runs, 2025-02-17 to 2026-07-08, surface plus lowest model level, area 30.8 N / 70.8 E / 8.4 N / 94.0 E, 0.4-degree grid. Reanalysis and missing-month substitution are prohibited.

What changed today

- The only request pending since 13 October (`0ffdede7`, June 2025 surface) returned **HTTP 404** when re-polled: the ADS server had deleted the result before anyone downloaded it. It was retired through a new reviewed path that keeps its ID on record (`retired_requests`); no other failure kind can be retired that way.
- The author **re-authorised submissions** of the remaining frozen jobs on 2026-10-20, one or two at a time (`submission_control.max_active = 2`). The worker polls every minute and downloads each result at once, so a finished result can no longer sit until it expires.
- June and July 2025 were submitted; both surface files are verified, both model-level files are running. Model-level files are slow: one earlier took 28.5 hours.

Request table

MONTH	FIELD GROUP	REQUEST ID	STATUS	BYTES	SHA-256 (PREFIX)
2025-02	surface	`30cb6db9`	VERIFIED_GRIB	10,454,784	353a103e97ec
2025-02	model level	`a0e3264d`	VERIFIED_GRIB	7,805,791	c8fbc18b64b4
2025-03	surface	`679e52de`	VERIFIED_GRIB	27,008,192	cddfa21527dd
2025-03	model level	`2e4f2f0c`	VERIFIED_GRIB	20,194,737	b1d310ab7a3c
2025-04	surface	`d0f52b0c`	VERIFIED_GRIB	26,136,960	5b19907596dd
2025-04	model level	`3ba0572d`	VERIFIED_GRIB	19,511,974	e703d43b301d
2025-05	surface	`1d4a865b`	VERIFIED_GRIB	27,008,192	025949192a64
2025-05	model level	`819b3723`	VERIFIED_GRIB	20,106,460	7783be665634
2025-06	surface	`d07238c3`	VERIFIED_GRIB	26,136,960	e3c9bd339c0f
2025-06	model level	`dc0ac7fb`	running	—	—
2025-07	surface	`c7bc38cb`	VERIFIED_GRIB	27,008,192	d5879fc1379e
2025-07	model level	`97ecdff2`	running	—	—
2025-06	surface (retired)	`0ffdede7`	HTTP 404, provider-deleted	—	—
2025-02-18	probe (not a plan job)	`33a7789e`	VERIFIED (134-byte GRIB1, PM2.5, 3×3 cells)	134	see `cams_archive_status.json`

Every verified file is re-hashed and decoded before use: forecast type, reference time, leads, valid times, fields, units, hybrid coefficients and grid are checked. Only months whose two halves are both verified enter station extraction.

Counts the brief asks for

	COUNT
Expected chunks	36

	COUNT
Verified	10
Running	2
Not yet submitted	24
Failed (unresolved)	0
Retired (provider-deleted, ID kept)	1
Duplicate requests	0
Paired months usable for the join	4 (Feb-May 2025; June and July await their model-level halves)
Eligible development dates in the join	103 (2025-02-19 to 2025-06-01)

External blocker: ADS queue time. The first protocol-eligible fold (fold_05) needs CAMS through 2025-11-16, i.e. 10 more verified files (`CAMS_STATION_MATCHING_REPORT.md`).

To continue acquisition (resumable, never duplicates, never submits beyond the frozen plan):

```
.venv-live\Scripts\python.exe -m src.research.calibration.cams_acquisition --worker-hours 24
```

Then, after new pairs verify: `python -m src.research.calibration.cams_baseline`, `python -m src.research.calibration.primary_dataset`, `python -m src.research.calibration.primary_experiments` (preflight) and `python -m scripts.build_cams_matching_report`.

CAMS-to-station matching report

Generated by `scripts/build_cams_matching_report.py` from the guarded development join; regenerate after each archive update. Machine-readable: `results/research/cams_station_matching_report.json`. The sealed final test (10 July to 7 October 2026) is outside the join by construction.

Current join

ITEM	VALUE
Station-days	18,540 (180 stations, 100 cities)
Target dates	103 (2025-02-19 to 2025-06-01)
Exact targets	11,634
Censored Severe targets (lower bound only)	7
Rejected targets kept visible	6,899
Temporal availability violations	**0**
Plan dates without CAMS yet	403
Join hash	`e9ecca99412bf916`

How a station is matched

- **Space**: nearest grid centre (station latitude and longitude each rounded to the 0.4-degree lattice); the cell must exist exactly once in the file. Grid: CAMS operational 0.4-degree grid.
- **Time**: native 3-hourly leads interpolated to hourly within the acquired leads; no extrapolation. Forecasts are 00 UTC runs issued the day before the target; observations obey the 20:00 IST cutoff.
- **Coast**: none: a coastal station can fall in a cell that is mostly sea; not yet audited.

Distance to the first usable fold

The protocol needs 120 valid training days per station. In the current join a station has at most 93 (median 75), and 63% of station-days have a valid target. Expanding folds grow by 30 days, so early folds cannot qualify whatever the archive holds.

FOLD	TRAINING DAYS	TRAINING ENDS	TEST WINDOW	CAMS NEEDED THROUGH	EXPECTED VALID DAYS PER STATION	CAN QUALIFY
fold_01	90	2025-05-19	2025-06-21 to 2025-07-20	2025-07-19	57	no
fold_02	120	2025-06-18	2025-07-21 to 2025-08-19	2025-08-18	75	no

FOLD	TRAINING DAYS	TRAINING ENDS	TEST WINDOW	CAMS NEEDED THROUGH	EXPECTED VALID DAYS PER STATION	CAN QUALIFY
fold_03	150	2025-07-18	2025-08-20 to 2025-09-18	2025-09-17	94	no
fold_04	180	2025-08-17	2025-09-19 to 2025-10-18	2025-10-17	113	no
fold_05	210	2025-09-16	2025-10-19 to 2025-11-17	2025-11-16	132	yes
fold_06	240	2025-10-16	2025-11-18 to 2025-12-17	2025-12-16	151	yes
fold_07	270	2025-11-15	2025-12-18 to 2026-01-16	2026-01-15	170	yes
fold_08	300	2025-12-15	2026-01-17 to 2026-02-15	2026-02-14	188	yes
fold_09	330	2026-01-14	2026-02-16 to 2026-03-17	2026-03-16	207	yes
fold_10	360	2026-02-13	2026-03-18 to 2026-04-16	2026-04-15	226	yes
fold_11	390	2026-03-15	2026-04-17 to 2026-05-16	2026-05-15	245	yes
fold_12	420	2026-04-14	2026-05-17 to 2026-06-15	2026-06-14	264	yes

First fold that can qualify: fold_05, which needs CAMS through 2025-11-16. 10 archive files up to that month are not yet verified. This is an expectation from the average valid share, not a guarantee for every station.

Limitations and future work

Checkpoint: 20 October 2026, commit after `9475d68` on `research/station-calibration`. Supersedes the 13 October checkpoint (preserved in git history). Final test sealed.

Categories: RESOLVED · PARTIALLY RESOLVED · BLOCKED BY EXTERNAL DATA · REQUIRES LONGER PROSPECTIVE RECORD · REQUIRES RESEARCH APPROVAL · REQUIRES FINAL-TEST AUTHORIZATION.

#	ORIGINAL PROBLEM	WHAT WAS IMPLEMENTED	EVIDENCE	CURRENT LIMITATION	SCIENTIFIC IMPACT	STATE	NEXT ACTION
1	Measured targets: QC, Severe censoring, target support	Locked v3.1 measured dataset; Severe kept as lower-bound censored	`measured_dataset.json` (179 stations, 100 cities, 383 censored Severe days)	Three-species target (PM2.5, PM10, O3), not the full bulletin AQI	Target is CPCB-style, not official	RESOLVED	Never redefine retrospectively
2	Operational CAMS archive coverage	Author re-authorised submissions on 2026-10-20; reviewed path for the expired June result; worker polls every minute and downloads at once	`cams_development_acquisition.json`: 10 of 36 files verified, more in flight	Model-level files can take over a day each; the laptop must stay on	No primary experiment until enough months arrive	BLOCKED BY EXTERNAL DATA (in progress)	Keep the worker running until all 36 files verify
3	Primary comparison, selection and lock	Guarded runner, preflight and join rebuilt reproducibly (identical hashes)	`CAMS_STATION_MATCHING_REPORT.md`: first eligible fold is fold_05, needing CAMS through 2025-11-16 (10 more files)	No eligible fold; no fit; no lock	No CAMS-correction claim possible	BLOCKED BY EXTERNAL DATA	Run the declared comparison once fold_05 is eligible
4	Severe-event forecasting	Bound-aware exploratory comparison (non-CAMS)	`UNCERTAINTY_EXTENSION_REPORT.md`, `DEVELOPMENT_EXTENSION_AUDIT.md`	No tail learner dominates; primary evaluation pending	Tail behaviour unknown for CAMS correction	PARTIALLY RESOLVED	Primary severity evaluation with winter folds
5	Uncertainty and warning costs	Exploratory calibration and operating points; production shows no interval	`WARNING_POLICY_EXTENSION_REPORT.md`; live page says "not yet available"	Marginal coverage only; polluted-category coverage poor in pilot	No interval or policy may be deployed	PARTIALLY RESOLVED	Primary Mondrian/CQR evaluation; two-sided cost function
6	Station survivorship and geography	Early-frozen cohort sensitivity	`STATION_SURVIVORSHIP_SENSITIVITY.md`	Retired monitors were never acquired; network not nationally representative	No nationwide generalisation claim	PARTIALLY RESOLVED	Disclose; leave-city/cluster-out on CAMS data
7	Coastal grid cells	Matching rule documented	`CAMS_STATION_MATCHING_REPORT.md`	Nearest 0.4-degree cell may be mostly sea for coastal stations; not audited	Possible bias at coastal stations	PARTIALLY RESOLVED	Land-sea mask audit before the primary run
8	Official CPCB feed	Connector, honest FAILED status	`DATA_SOURCE_REGISTER.md`: 0 successful, 187 failed ingestions; on 2026-10-20 the host answered but returned HTTP 502 after ~60 s	No official CPCB data in the project	Targets come from OpenAQ's copy of CPCB monitors	BLOCKED BY EXTERNAL DATA	Retry; never relabel another source as CPCB

#	ORIGINAL PROBLEM	WHAT WAS IMPLEMENTED	EVIDENCE	CURRENT LIMITATION	SCIENTIFIC IMPACT	STATE	NEXT ACTION
9	Public track record	Errors published only for groups with ≥30 checked forecasts over ≥7 days (deployed 2026-10-20)	`src/nationwide/store.py`; live page: 48 forecasts on 3 days, nothing scored	Too few verified days	No live accuracy claim	REQUIRES LONGER PROSPECTIVE RECORD	Wait; never backfill
10	Prospective research ledger	Fixed 20:00 IST issuance since 2026-10-11	`prospective_audit.json`: 108 forecasts, 9 target days, 2 verified days, 0 availability violations	30-day gate far off	No prospective skill claim	REQUIRES LONGER PROSPECTIVE RECORD	Keep issuing; score after ≥30 verified days
11	Stale data shown as current	Summaries built mid-refresh cached ≤30 s (deployed 2026-10-20)	`tests/test_nationwide.py` reproduces the overlap	First uncached visit takes ~20 s	None on accuracy	RESOLVED	Monitor
12	Literature and novelty	Identifier verification (48 of 56 verified), logged Crossref search, 28-study matrix, gap and claim matrices	`FINAL_NOVELTY_AND_PRIOR_ART_AUDIT.md`	One database, top-20 hits, title-only reads for some; no human acceptance	Novelty is a design combination, not yet a result	PARTIALLY RESOLVED	Full texts of title-only studies; human acceptance
13	Public-impact evaluation	Task-based protocol with pre-registered thresholds	`PUBLIC_IMPACT_EVALUATION_PLAN.md`, `docs/USER_STUDY.md`	No participant; nothing collected	No comprehension or health claim	REQUIRES RESEARCH APPROVAL	Ethics approval, then run
14	Tamil and Hindi guidance	Disabled pending named-reviewer approval	`docs/I18N_REVIEW.md`	English only	Reach limited	REQUIRES RESEARCH APPROVAL	Reviewer sign-off
15	Final-test evaluation	Code-level guards and tests (sealed rows dropped before loading; tuning refuses sealed dates)	`tests/test_primary_development_alignment.py`, `tests/test_primary_experiments.py`	No lock, so no READY gate	Final generalisation unknown	REQUIRES FINAL-TEST AUTHORIZATION	Request approval only after a committed lock and READY audit
16	Credentials pasted into chat earlier	Values kept out of code, logs and bundles; secret audit 0/0/0 on 2026-10-20	`results/release/secret_audit.json`	The keys and database password were exposed in a chat transcript	Security, not science	PARTIALLY RESOLVED	Author to rotate the three API keys and the database password

Final novelty and prior-art audit

Generated 2026-10-20T07:39:55Z by `scripts/build_novelty_audit.py`. Machine-readable: `results/research/literature_matrix.json`. This audit supersedes the bounded `NOVELTY_AUDIT.md` and builds on `LITERATURE_FULLTEXT_REVIEW.md`; nothing there is discarded.

Verdict

AirGuard's individual methods are all established: station-level CAMS correction, gradient boosting and ridge models, rolling-origin and spatial validation, conformal intervals, cost-loss warnings and operational forecast ledgers each have prior art below, and India already runs operational early-warning systems. **No first-ever or state-of-the-art claim is supported.**

What was not found among the reviewed studies is the combination: archived operational CAMS forecasts corrected at CPCB-network monitors across about 100 Indian cities, under a code-enforced issue-time contract, judged by temporal and unseen-station transfer, with the open Severe band treated as censored and with coverage and under-warning reported by pollution category. That combination is a defensible *research design*. It becomes a *contribution* only when the primary CAMS experiment has run; today it has not (archive 9 of 36 files, no eligible fold).

How the literature was checked

- **Identifier verification** (`scripts/literature_audit.py verify`, 2026-10-20T07:37:16Z): 56 records from the paper bibliography and the full-text review. 48 verified against Crossref, arXiv or the official landing page; 6 are official web resources that resolve; 2 could not be machine-verified (the 2014 CPCB AQI report, whose official PDF URL returns a web page, and Holm 1979, whose JSTOR page blocks automated access). Both are standard references; they are flagged, not removed, because the paper is a frozen snapshot.
- **Logged search** (`scripts/literature_audit.py search`, 2026-10-20T07:37:45Z): 10 fixed queries against Crossref, top 20 journal articles from 2018 each, 190 unique records. Titles were screened for air quality plus a relevant method or region; 13 studies were judged relevant and added below at the depth actually read.
- **Limits**: One bibliographic database; top 20 relevance-ranked hits per query; journal articles only, so conference papers and preprints are covered only through the existing review. This is a targeted, reproducible audit, not a systematic review; no independent human has accepted it.

Literature matrix

Design facts are given only where the text was read. "not established" means the audit did not read enough to say.

STUDY	VENUE / DOI	READ	DATA AND GEOGRAPHY	METHOD / TASK	EVALUATION, SPATIAL HOLDOUT	INPUTS AT ISSUE TIME	UNCERTAINTY, EXTREMES, WARNINGS	RELATION TO AIRGUARD
Riccio and Chianese (2024): Technical note: Accurate, reliable, and high-resolution air quality predictions by improving the Copernicus At	10.5194/acp-24-1673-2024 (VERIFIED_DOI)	FULLTEXT_SELECTED_SECTIONS	Italy; 750-station network; 2022.	Daily PM10/PM2.5; maximum 8-hour O3 and 1-hour NO2; EMOS then INLA-SPDE.	668 training/82 stratified held-out stations; next-day repetition using previous three days.	CAMS available 08 UTC. Main 2.2 says ECMWF operational weather; supplement S1 lists ERA5 precipitation, wind and boundary-layer height. Rece	Predictive distributions, PIT, intervals and exceedance diagnostics; cohort advice not established.	Direct overlap with CAMS station correction and spatial uncertainty. Weather provenance discrepancy prevents treating this as proof of issue-time validity.
Petetin et al. (2022): Model output statistics (MOS) applied to Copernicus Atmospheric Monitoring Service (CAMS) O3 forecasts: trade-	10.5194/acp-22-11603-2022 (VERIFIED_DOI)	FULLTEXT_SELECTED_SECTIONS	Iberian domain and southwestern France; 455 stations; 2018-2019.	O3 MOS; persistence, moving averages, quantile mapping, Kalman, analogs and boosting; 1-4-day leads.	Expanding training starts January 2018; only accumulated past observations; matched complete forecast-observation pairs.	ECMWF HRES forecast weather primary; ERA5 sensitivity separate. Previous-day O3 reporting availability explicitly questioned and ablated.	Continuous versus hits/misses/false alarms; boosting can suppress peaks; tail weighting changes trade-offs.	Established station CAMS correction. Continuous improvement does not establish extreme-event warning improvement; no cohort health-benefit result.
Casciaro, Cavaiola and Mazzino (2022): Calibrating the CAMS European multi-model air quality forecasts for regional air pollution monitoring	no DOI (VERIFIED_ARXIV)	FULLTEXT_SELECTED_SECTIONS	Liguria, northwestern Italy; 2019-2020.	Dynamic gamma-EMOS calibration of nine CAMS members; PM10, PM2.5, O3, NO2 and CO.	2019 training/cross-validation; 2020 evaluation; additional rolling 40-day calibration.	CAMS 08 UTC; dynamic persistence uses 09 UTC observations and previous-day values; WRF forecast weather. Ingestion receipts not audited.	Gamma predictive distribution, CRPS and PIT-based reliability; no cohort advice established.	Direct overlap with observation/weather-conditioned correction; static calibration does not uniformly beat persistence. No India CPCB-AQI experiment.
Colette et al. (2025): Copernicus Atmosphere Monitoring Service - Regional Air Quality Production System v1.0	10.5194/gmd-18-6835-2025 (VERIFIED_DOI)	FULLTEXT_SELECTED_SECTIONS	Thousands of European observation sites; PM10, PM2.5, NO2 and O3; 96-hour forecasts.	European regional operational system; CAMS-MOS random forest uses ensemble concentrations and forecast meteorology.	MOS trains daily on previous three days. System forecasts evaluated against available stations; analyses reserve approximately one-third out	Forecast, analysis and reanalysis products differentiated. Daily MOS operational in ADS; AirGuard receipt provenance not established.	Ensemble dispersion; threshold-detection diagnostics; no India cohort guarantee.	Operational precedent for station correction. System description is not an isolated MOS ablation study; reanalysis diagnostic figures are not forecast comparisons.
Jena et al. (2021): Performance of high resolution (400 m) PM2.5 forecast over Delhi	10.1038/s41598-021-83467-8 (VERIFIED_DOI)	PRIMARY_INDEXED_FULLTEXT_SECTIONS	Delhi/NCR; 37 CPCB/DPCC/IITM stations; October 2019-February 2020.	Daily 72-hour, 400 m WRF-Chem PM2.5 forecast with AOD/surface assimilation and fire emissions.	Operational seasonal evaluation by lead; geographic station holdout not established.	Near-real-time observations/fire inputs reported; exact issue/receipt audit not established.	Concentration and AQI-category detection/false alarms; Severe forecast difficulty. Calibrated intervals not established.	Indian operational forecasting precedent; distinct from CAMS residual correction. Evaluation-site assimilation independence requires further inspection.
Natarajan et al. (2024): Optimized machine learning model for air quality index prediction in major cities in India	10.1038/s41598-024-54807-1 (VERIFIED_DOI)	PRIMARY_INDEXED_FULLTEXT_SECTIONS	India Kaggle 2015-2020 data; six-city evaluation.	Grey-wolf feature selection and decision-tree AQI regression.	Null removal and SMOTE described; chronological/spatial isolation not established in inspected text.	Pollutant-to-AQI inputs described; future forecast horizon and issue cutoff not established.	Deterministic regression metrics; intervals and cohort advice not established.	Indian AQI ML precedent. Reported accuracy cannot establish next-day operational forecast skill or absence of leakage without a timestamp/split audit.
Govardhan et al. (2024): Decision Support System version 1.0 (DSS v1.0) for air quality management in Delhi, India	10.5194/gmd-17-2617-2024 (VERIFIED_DOI)	FULLTEXT_SELECTED_SECTIONS	Delhi/NCR; 39 evaluation locations; October 2021-February 2022.	WRF-Chem five-day PM2.5/source-apportionment/emission-scenario forecasts.	Daily leads 1-5 evaluated separately by post-monsoon/winter season; no leave-city-out experiment established.	IITM-GFS forecast, emission inventories, VIIRS fires, CPCB/satellite assimilation; exact receipt timestamps not audited.	Category detection, misses and false alarms; poorer/severe episodes underpredicted; source scenarios support management.	Indian operational advisory precedent. Source/emission scenarios do not identify an individual health benefit; distinct from calibrated CAMS station residuals.
Singh et al. (2026): Estimation of surface PM2.5 over the Indo-Gangetic Basin using MERRA-2 reanalysis and machine learning	10.1038/s41598-026-37934-9 (VERIFIED_DOI)	PUBLISHER_MANUSCRIPT_AND_INDEXED_SECTIONS	183 CPCB stations, 2014-2023; Delhi, Kanpur, Lucknow and Patna modeling.	RF, Extra Trees, LightGBM and stacking estimate daily city-background PM2.5.	75/25 train/test; tenfold K-fold comparison; chronological/spatial separation not established.	Same-day MERRA-2 aerosol and weather reanalysis; not issue-time operational forecasts.	Deterministic errors/seasonal analysis; calibrated intervals/cohort advice not established.	Indian measured-data correction precedent; reanalysis estimation cannot substitute the required archived CAMS forecast baseline.

STUDY	VENUE / DOI	READ	DATA AND GEOGRAPHY	METHOD / TASK	EVALUATION, SPATIAL HOLDOUT	INPUTS AT ISSUE TIME	UNCERTAINTY, EXTREMES, WARNINGS	RELATION TO AIRGUARD
Pande et al. (2024): Daily scale air quality index forecasting using bidirectional recurrent neural networks: Case study of Delhi,	10.1016/j.envpol.2024.124040 (VERIFIED_DOI)	ABSTRACT_AND_PRIMARY_INDEXED_BODY_SECTIONS	Delhi; 3318 daily points; seven pollutant inputs. Dataset section ends December 2022; results/conclusion end January 2023.	Daily AQI using LSTM, Bi-LSTM, Bi-RNN and KRR.	Training/testing reported; exact chronology, tuning isolation and forecast lead not established.	Input pollutant measurements stated; lag/receipt cutoff validity unresolved.	Deterministic metrics; uncertainty and cohort advice not established.	Indian AQI forecast prior art; full methods needed before claiming comparable forecast horizons or leakage protection.
Romano, Patterson and Candes (2019): Conformalized Quantile Regression	no DOI (VERIFIED_ARXIV)	FULLTEXT_SELECTED_SECTIONS	Synthetic and 11 generic regression benchmarks; not an air-quality study.	Quantile regression plus held-out conformal interval calibration.	Disjoint proper training/calibration; experiments 40/40/20 train/calibration/test with repeated random splits.	Preprocessing fitted on proper training; operational issue-time availability outside scope.	Finite-sample rank gives marginal coverage under exchangeability; adaptive interval width.	Established interval method. Does not automatically guarantee dependent daily, city, vulnerability-cohort or Severe-conditional coverage.
Angelopoulos, Bates, Fisch, Lei and Schuster (2024): Conformal Risk Control	no DOI (VERIFIED_ARXIV)	FULLTEXT_SELECTED_SECTIONS	Segmentation, multilabel vision, hierarchical classification and question-answering examples; no India air-quality data.	Post-process fixed models to control expected bounded monotone loss.	Separate observed calibration losses and unseen test loss; exchangeability required.	Predict then evaluate unseen point; operational exposure timestamps outside scope.	Expected-risk bound with finite-sample correction; not a high-probability conditional risk bound.	Under-warning loss can be related only after assumptions/monotonicity are justified. Daily dependence and cohort health risk remain unvalidated.
Angelopoulos, Barber and Bates (2024): Online conformal prediction with decaying step sizes	no DOI (URL_REACHABLE)	FULLTEXT_SELECTED_SECTIONS	Australian Elec2 demand and ImageNet sequences; no air-quality experiment.	Predict-then-observe interval updates with decaying step sizes.	Time-ordered online prediction; Elec2 even/odd sequence/holdout; ImageNet sequence/holdout.	Base prediction uses past data; score updates after outcome arrival.	Bounded-score arbitrary-sequence long-run coverage; IID quantile convergence under stated conditions.	Long-run historical coverage is distinct from next-day/cohort-conditional coverage. Static split-conformal calibration does not implement this update method.
Adjei et al. (2026): Conformal PM2.5 Mapping Under Spatial Covariate Shift: Satellite-Reanalysis Fusion for Africa's Green Industri	no DOI (VERIFIED_ARXIV)	FULLTEXT_SELECTED_SECTIONS	404 locations, 29 African countries; OpenAQ 2017-2022.	Satellite/reanalysis PM2.5 mapping with LightGBM and split conformal intervals.	Five location-grouped folds; 80/20 proper train/calibration; random inner tuning described.	ERA5, MERRA-2 and satellite inputs; location-year standardization described; not issue-time forecasting.	Regional coverage degradation and reliability flags; no demonstrated clinical intervention benefit.	Relevant spatial-calibration diagnostics. Inner splits/group normalization need independent audit; stated spatial leakage resistance does not establish temporal availability.
WHO Regional Office for Europe (2024): Personal-level actions to reduce air pollution exposure in the WHO European Region	no DOI (URL_REACHABLE)	OFFICIAL_OVERVIEW_ONLY	WHO European Region; transfer to India requires local assessment.	Guidance reviewing personal exposure-reduction actions and implementation context.	Not predictive-model evaluation.	Guidance, not a forecast-input dataset.	Effectiveness, costs and social feasibility considered at overview level.	Supports reviewing feasible exposure advice, not AirGuard medical coefficients, 10%/5% policy targets or demonstrated health benefit.
US EPA / AirNow (undated webpages inspected 2026-10-12): AQI Basics; Asthma and Heart Disease	no DOI (URL_REACHABLE)	OFFICIAL_WEBPAGES	United States; US AQI categories and thresholds.	Public AQI and sensitive-group guidance.	No model split or AirGuard intervention evaluation.	Guidance, not predictive input provenance.	Pollutant/category-specific sensitive-group advice.	Vulnerability-aware air-quality guidance already exists. US thresholds must not replace CPCB thresholds; no individualized disease probabilities or AirGuard clinical benefits follow.
Munteanu et al. (2026): Evaluating Graph-Centrality Context for Station-Level Correction of CAMS Air Quality Reanalysis in Lombardy fr	10.3390/s26185844 (VERIFIED_DOI (Crossref search record))	ABSTRACT_ONLY	CAMS European reanalysis vs ARPA Lombardia stations, Italy, 2018-2021	graph-centrality features with GTWR, XGBoost, random forest	not established	not established	not established	Station-level CAMS correction exists; it uses reanalysis, not archived forecasts available at issue time.
Sarafian et al. (2025): Correction of CAMS PM10 Reanalysis Improves AI-Based Dust Event Forecast	10.3390/rs17020222 (VERIFIED_DOI (Crossref search record))	ABSTRACT_ONLY	CAMS PM10 reanalysis vs in-situ stations; dust events	ML correction of CAMS PM10 fields with in-situ data	not established	not established	targets rare dust events	ML correction of CAMS against stations for rare events; reanalysis input, so not an issue-time forecast test.

STUDY	VENUE / DOI	READ	DATA AND GEOGRAPHY	METHOD / TASK	EVALUATION, SPATIAL HOLDOUT	INPUTS AT ISSUE TIME	UNCERTAINTY, EXTREMES, WARNINGS	RELATION TO AIRGUARD
Stortini et al. (2020): Operational Forecast and Daily Assessment of the Air Quality in Italy: A Copernicus-CAMS Downstream Service	10.3390/atmos11050447 (VERIFIED_DOI (Crossref search record))	ABSTRACT_ONLY	Italy, national domain	CAMS-downstream national forecasting and assessment chain	not established	not established	not established	Operational CAMS-downstream services exist; AirGuard is not the first operational CAMS-based forecaster.
June et al. (2021): Operational bias correction for PM2.5 using the AIRPACT air quality forecast system in the Pacific Northwest	10.1080/10962247.2020.1856216 (VERIFIED_DOI (Crossref search record))	TITLE_ONLY	Pacific Northwest, USA	operational PM2.5 bias correction (details not read)	not established	not established	not established	Operational station bias correction of a chemical-transport forecast is established practice.
Gualtieri et al. (2025): Assessing capability of Copernicus Atmosphere Monitoring Service to forecast PM2.5 and PM10 hourly concentrati	10.1016/j.apr.2025.102567 (VERIFIED_DOI (Crossref search record))	TITLE_ONLY	a European hotspot	evaluation of CAMS hourly PM forecasts	not established	not established	not established	CAMS forecast skill against stations is routinely evaluated outside India.
Kaffashzadeh and Kazemi (2024): Assessment of surface ozone products from downscaled CAMS reanalysis and CAMS daily forecast using urban air q	10.5194/gmd-17-4155-2024 (VERIFIED_DOI (Crossref search record))	ABSTRACT_ONLY	27 urban stations, Iran, 2020	spectral decomposition of CAMS reanalysis and daily-forecast ozone	not established	not established	not established	CAMS surface-ozone mismatch at urban stations is documented elsewhere; consistent with AirGuard's large ozone correction factors.
Ghude et al. (2020): Evaluation of PM2.5 Forecast using Chemical Data Assimilation in the WRF-Chem Model: A Novel Initiative Under	10.18520/cs/v118/i11/1803-1815 (VERIFIED_DOI (Crossref search record))	TITLE_ONLY	Delhi, India	WRF-Chem with chemical data assimilation	not established	not established	not established	India already has an operational national early-warning system for Delhi; AirGuard does not replace or rival it.
Kalita et al. (2023): Forecasting of an unusual dust event over western India by the Air Quality Early Warning System	10.1016/j.atmosenv.2023.120013 (VERIFIED_DOI (Crossref search record))	TITLE_ONLY	western India	not established	not established	not established	single dust event	Indian operational extreme-event forecasting exists for specific events.
Lawrence et al. (2025): Evaluating machine learning models and imputation strategies for Air Quality Index forecasting in urban India	10.1007/s10661-025-14700-4 (VERIFIED_DOI (Crossref search record))	TITLE_ONLY	urban India	ML models and imputation for AQI forecasting	not established	not established	not established	Indian AQI forecasting with ML is a crowded area; model choice alone is not novel.
Dar et al. (2025): Spatiotemporal clustering and multivariate forecasting of air quality index across Indian cities using machine	10.1016/j.fraope.2025.100435 (VERIFIED_DOI (Crossref search record))	TITLE_ONLY	multiple Indian cities	clustering plus ML/DL forecasting	not established	not established	not established	Multi-city Indian AQI forecasting and clustering exist.

STUDY	VENUE / DOI	READ	DATA AND GEOGRAPHY	METHOD / TASK	EVALUATION, SPATIAL HOLDOUT	INPUTS AT ISSUE TIME	UNCERTAINTY, EXTREMES, WARNINGS	RELATION TO AIRGUARD
B. et al. (2026): Integrated air quality monitoring and PM2.5 forecasting in India: a comparative study of machine learning and	10.1007/s11869-026-01973-y (VERIFIED_DOI (Crossref search record))	TITLE_ONLY	India	ML vs DL comparison	not established	not established	not established	Further Indian ML/DL comparison; whether issue-time availability was audited is not established.
Murad et al. (2021): Probabilistic Deep Learning to Quantify Uncertainty in Air Quality Forecasting	10.3390/s21238009 (VERIFIED_DOI (Crossref search record))	ABSTRACT_ONLY	not established	probabilistic deep learning	not established	not established	predictive uncertainty, reliability of confidence estimates	Uncertainty-aware air-quality forecasting is established; severity-conditioned coverage is the narrower question.
Wang et al. (2018): Application of a novel early warning system based on fuzzy time series in urban air quality forecasting in Chi	10.1016/j.asoc.2018.07.030 (VERIFIED_DOI (Crossref search record))	TITLE_ONLY	Chinese cities	not established	not established	not established	early-warning system	Air-quality early-warning systems with ML forecasting are established.

Gap matrix

AIRGUARD ELEMENT	PRIOR-ART STATUS	CLOSEST STUDIES	WHAT WOULD BE AIRGUARD'S	EVIDENCE TODAY
Station-level correction of CAMS forecasts	ESTABLISHED	petetin_2022, bertrand2023, riccio_chianese_2024, casciaro_2022, munteanu_2026, june_2021	Applying it to CPCB-network monitors in Indian cities was not found among the reviewed studies.	PENDING: no primary CAMS comparison has run (archive at 9/36 files).
Correction trained only on forecasts available at issue time (archived operational runs, 20:00 IST contract)	PARTIALLY ESTABLISHED	petetin_2022 and riccio_chianese_2024 use operational CAMS; munteanu_2026 and sarafian_2025 use reanalysis	An explicit, code-enforced availability contract with a violation count is AirGuard's design.	Guard implemented; 0 violations in the 18,540-row join; publication times are nominal, not observed.
Temporal and geographic transfer evaluation (rolling origin, unseen stations and cities)	ESTABLISHED METHOD	bergmeir2012, roberts2017, meyer2018, garciacrespi2026, riccio_chianese_2024 (held-out stations)	Application to ~100 Indian cities.	Done for the non-CAMS paper; pending for CAMS.
Severe AQI treated as lower-bound censored, never as an exact 401	NOT FOUND	none of the reviewed studies treats the open Severe band as censored	Possible contribution, narrow.	Implemented; 383 censored Severe station-days in the measured set; primary CAMS evaluation pending.
Coverage audited by pollution category (Mondrian / severity-conditioned)	PARTIALLY ESTABLISHED	romano2019, xu2021, gibbs2021, angelopoulos2024, adjei_2026, murad_2021	Reporting that marginal coverage hides poor coverage in polluted categories, for Indian AQI.	Exploratory, non-CAMS: 90% intervals covered 57-75% of Severe days (pilot); no guarantee claimed.
Under-warning versus over-warning trade-off for advisories	ESTABLISHED CONCEPT	murphy1977 (cost-loss), wang_2018 (early warning), angelopoulos2024 (risk control)	Two-sided warning losses per cohort are AirGuard's framing.	Exploratory only; no policy is called optimal.
Immutable forecast ledger and prospective verification	ESTABLISHED PRACTICE	operational services such as stortini_2020, colette_2025, ghude_2020	Engineering, not novelty.	Running since 2026-10-11: 108 fixed-time forecasts, 10 verified outcomes; no skill claim.
Separation of raw, corrected and experimental forecasts in the public product	ESTABLISHED PRACTICE	operational services label model output	Engineering and communication, not novelty.	Live: modelled labels, no deployed correction or interval.

Claim-evidence matrix

CLAIM	STATUS	EVIDENCE
AirGuard forecasts next-day AQI across Indian cities from CAMS-based inputs	**SUPPORTED (as a service)**	live site; docs/DATA_SOURCES.md
The forecast is accurate / beats CAMS or persistence	**NOT SUPPORTED**	no primary CAMS experiment; live track record has 3 verified days; 2026 backtest of the 1.0 model did not beat persistence
Station-calibrated CAMS correction improves Indian forecasts	**NOT YET TESTED**	CAMS archive 9/36; MODEL_SELECTION_REPORT.md: NOT READY
Weather Ridge beats climatology on held-out dates (non-CAMS paper)	**SUPPORTED**	paper snapshot; Strict Ridge MAE 8.96 lower than climatology, 95% block interval [-12.49, -5.61]
History Ridge beats persistence	**NOT ESTABLISHED**	PROGRESS.md; MODEL_SELECTION_REPORT.md
Intervals are calibrated in polluted categories	**NOT SUPPORTED**	pilot: Severe coverage 57-75% for nominal 90%
Any warning policy is optimal	**NOT SUPPORTED**	WARNING_POLICY_EXTENSION_REPORT.md
AirGuard is the first / state of the art	**NOT SUPPORTED**	operational CAMS-downstream and Indian early-warning systems exist (this audit)
AirGuard improves health outcomes	**NOT SUPPORTED**	no outcome study; HAI is a heuristic, not clinically validated
Severe days are handled without inventing exact values	**SUPPORTED (method)**	censoring in src/nationwide/aqi.py and the research metrics

What may be said

- "Among the studies reviewed, none evaluated archived operational CAMS forecasts corrected at CPCB-network monitors across Indian cities under an explicit issue-time contract with censored Severe outcomes." (scoped, true today)
- Not: "first", "novel method", "state of the art", "validated", "improves health", "guaranteed coverage".

Remaining actions

- Run the primary CAMS experiment when the archive allows an eligible fold; only then can the gap become a contribution.
- Read the full texts of the title-only additions (Ghude 2020, Kalita 2023, June 2021, Lawrence 2025) before citing their designs.
- Independent human acceptance of this audit and of the frozen paper's bibliography.

Data source register

Generated 2026-10-20T07:31:09Z by `scripts/build_data_source_register.py` from committed evidence and read-only database counts. Machine-readable copy: `results/research/data_source_manifest.json`. Licences and attribution: `docs/LICENSES.md`; provider quotas: `docs/DATA_SOURCES.md`.

Every source carries one of five kinds and none stands in for another: **OBSERVED**, **HISTORICAL_REANALYSIS** (none is used), **ARCHIVED_OPERATIONAL_FORECAST**, **LIVE_FORECAST**, **MODELLED_OUTPUT**.

SOURCE	KIND	PROVIDER	COVERAGE	ISSUE-TIME STATUS	EVIDENCE
`A_B_station_observations`	OBSERVED	OpenAQ v3, upstream provider CPCB (Central Pollution Control Board monitoring network)	['2025-02-19', '2026-10-07']	see notes	`results/research/measured_datase` `results/research/data_manifests/`
`A_direct_cpcb_feed`	OBSERVED	CPCB via data.gov.in	-	UNAVAILABLE: no successful response recorded	`docs/airguard2/live_data_impleme` `airguard.ingestion_runs`
`C_weather_archive`	ARCHIVED_OPERATIONAL_FORECAST	Open-Meteo (NOAA GFS, model gfs_seamless)	['2025-02-01', '2026-10-08']	valid	`results/research/data_manifests/`

SOURCE	KIND	PROVIDER	COVERAGE	ISSUE-TIME STATUS	EVIDENCE
`D_cams_operational_archive`	ARCHIVED_OPERATIONAL_FORECAST	Copernicus Atmosphere Monitoring Service (ECMWF), via the Atmosphere Data Store	['2025-02-17', '2025-05-31']	valid	`results/research/cams_developmen` `results/research/cams_developmen`
`D_cams_short_lead_proxy`	MODELLED_OUTPUT	CAMS global via Open-Meteo	['2025-02-01', '2026-10-08']	NOT valid	`results/research/data_manifests/`
`E_station_metadata`	OBSERVED	OpenAQ locations, with versioned AirGuard corrections	-	see notes	`results/research/data_manifests/` `results/research/data_manifests/`
`F_legacy_aqi`	OBSERVED	Kaggle export of CPCB data (2015-2020)	-	see notes	`data/raw/PROVENANCE.json`
`G_live_inputs`	LIVE_FORECAST	Open-Meteo (CAMS global air quality, GFS weather), OpenAQ, NASA FIRMS	-	see notes	`docs/DATA_SOURCES.md` , `database:`
`H_prospective_records`	LIVE_FORECAST	AirGuard (issued forecasts) and OpenAQ (later outcomes)	-	see notes	`database: airguard.forecast_runs` `verification tables`
`satellite_fire`	OBSERVED	NASA FIRMS	['2025-02-01', '2026-10-08']	see notes	`results/research/data_manifests/`

What the numbers say

- **Station observations** come from CPCB-network monitors **through OpenAQ**: 179 usable stations in 100 Indian cities, 2025-02-19 to 2026-10-07. This is not official CPCB data and the targets are AirGuard's CPCB-style daily AQI from PM2.5, PM10 and ozone, not the bulletin AQI.
- Missing data: of 107,280 station-days, 42,481 (39.6%) have no valid target. Of 6,331,429 raw hourly rows, 851,125 fall in under-covered hours.
- **The official CPCB feed** (data.gov.in) has never returned data to this project: 0 successful and 187 failed recorded ingestions.
- **Archived operational CAMS**: 9 of 36 planned ADS files verified, covering 104 initialisation dates with both surface and model-level fields (2025-02-17 to 2025-05-31). Current join: 18,540 station-days, 180 stations, 103 dates. Reanalysis substitution is prohibited.
- **The Open-Meteo CAMS past-hours series is not an archived forecast**: its values come from runs initialised after the issue time, so it is kept as exploratory modelled output only.
- **Weather predictors** use Open-Meteo's previous-runs archive at a two-day lead, which is older than what an operator would have had, so it handicaps rather than flatters the weather features. Boundary-layer height is not archived at any previous-run lead.
- **Fire detections** are archived but withheld from the strict models: acquisition time does not prove publication time.
- **Prospective records**: 108 fixed-time research forecasts over 9 target days with 10 verified outcomes; 532 on-demand nationwide forecasts with 77 station pairs over 3 verified target days.
- **Live inputs**: 5,353 recorded provider events.

Per-source details (endpoint, units, sampling, licence, quality control, issue-time rule) are in the JSON manifest.